Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Koolhaas Dairy

Physical address of dairy:

26805 Dodds RD	Escalon	San Joaquin	95320
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X207-X140-XX06-XXXX X207-X140-XX07-XXXX

B. OPERATORS

Koolhaas, David

Operator name: Koolhaas, David

Telephone no.: Landline: ; Cellular: (209) 606-3424

26805 Dodds RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

Koolhaas, Kenneth

Operator name: Koolhaas, Kenneth

Telephone no.: Landline: (209) 866-1983; Cellular:

26805 Dodds RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Vanderschaaf, Dodds Road LP

Legal owner name: Vanderschaaf, Dodds Road LP

Telephone no.: Landline: ; Cellular: (209) 606-3424

26805 Dodds RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	150	300	0	0
Number under roof	864	125	0	0	0	0
Maximum number	864	125	150	300	0	0
Average number	864	125	150	300	0	0
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 28,957.03 *tons per reporting period*

Total nitrogen from manure: 361,914.77 *lbs per reporting period*

After ammonia losses (30% loss applied): 253,340.34 *lbs per reporting period*

Total phosphorus from manure: 60,513.17 *lbs per reporting period*

Total potassium from manure: 168,302.21 *lbs per reporting period*

Total salt from manure: 435,558.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 38,070,000 *gallons*

Total nitrogen generated: 137,032.53 *lbs*

Total phosphorus generated: 17,054.72 *lbs*

Total potassium generated: 212,494.13 *lbs*

Total salt generated: 1,615,607.69 *lbs*

	38,070,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	38,070,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW 4	Ground water
IW 5	Ground water
IW 6	Ground water
IW 7	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/26/2023	Solid commercial fertilizer *Starter 4-0-0*	51.30 *ton*	As-is	0.1	4.000000	0.000000	0.000000	0.000000
07/20/2023	Solid commercial fertilizer *UN32 Corn*	6.65 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	8,360.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	8,360.00	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	90	90	2	process wastewater	X207-X140-XX06-XXXX
Field 2	75	75	2	process wastewater	X207-X140-XX06-XXXX
Field 3	35	35	2	process wastewater	X207-X140-XX06-XXXX
Field 4	55	55	2	both	X207-X140-XX06-XXXX
Field 5	35	35	2	process wastewater	X207-X140-XX07-XXXX
Field 6	62	62	2	manure	X207-X140-XX05-XXXX X207-X140-XX06-XXXX
Field 7	32	32	2	both	X207-X140-XX07-XXXX
Field 8	27	27	2	manure	X207-X140-XX07-XXXX
Totals for areas that were used for application	411	411	16		
Totals for areas that were not used for application					
Land application area totals	411	411	16		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 90 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,476.00 *ton*	As-is		61.1	6,340.00	1,320.00	10,400.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	16.40	207.95	43.30	341.12	1,467.31

Field 1

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 90 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	2,475.00 *ton*	As-is		67.2	4,000.00	740.00	3,310.00		5.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.50	220.00	40.70	182.05	966.94

Field 2

Field name: Field 2

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	922.50 *ton*	As-is		57.6	8,270.00	1,450.00	11,500.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	12.30	203.44	35.67	282.90	1,063.90

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	1,950.00 *ton*	As-is		64.2	4,830.00	830.00	4,000.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.00	251.16	43.16	208.00	1,284.50

Field 3

Field name: Field 3

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	546.00 *ton*	As-is		55.1	6,870.00	1,430.00	11,300.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	15.60	214.34	44.62	352.56	1,428.90

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	976.50 *ton*	As-is		71.2	3,720.00	750.00	3,470.00		7.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.90	207.58	41.85	193.63	1,275.99

Field 4

Field name: Field 4

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	621.50 *ton*	As-is		48.0	7,020.00	1,450.00	11,900.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	11.30	158.65	32.77	268.94	1,398.49

Field 4

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	1,413.50 *ton*	As-is		61.4	4,590.00	760.00	3,580.00		5.52

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	25.70	235.93	39.06	184.01	1,095.19

Field 5

Field name: Field 5

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	616.00 *ton*	As-is		69.3	4,640.00	850.00	6,600.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.60	163.33	29.92	232.32	1,156.28

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	1,001.00 *ton*	As-is		71.4	3,630.00	590.00	3,080.00		6.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.60	207.64	33.75	176.18	1,102.61

Field 6

Field name: Field 6

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 62 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,103.60 *ton*	As-is		61.4	4,010.00	810.00	6,260.00		8.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	17.80	142.76	28.84	222.86	1,107.57

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 62 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	1,661.60 *ton*	As-is		66.8	4,610.00	920.00	4,060.00		7.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.80	247.10	49.31	217.62	1,293.71

Field 7

Field name: Field 7

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	646.23 *ton*	As-is		72.6	3,510.00	750.00	7,440.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	20.19	141.77	30.29	300.50	1,117.74

Field 7

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	803.20 *ton*	As-is		58.8	5,400.00	1,080.00	4,200.00		4.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	25.10	271.08	54.22	210.84	984.48

Field 8

Field name: Field 8

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	510.30 *ton*	As-is		66.7	3,760.00	840.00	8,240.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	18.90	142.13	31.75	311.47	1,246.15

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	712.80 *ton*	As-is		65.0	4,590.00	860.00	3,510.00		6.33

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.40	242.35	45.41	185.33	1,169.78

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/10/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.37	18.40	161.10	1,536.65	2,970,000.00 *gal*
Application event totals		123.37	18.40	161.10	1,536.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.07	20.74	181.60	1,732.22	3,348,000.00 *gal*
Application event totals		139.07	20.74	181.60	1,732.22	

Field 1 - 06/22/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.55	0.00	0.00	125.94	6,468,000.00 *gal*
Application event totals		2.55	0.00	0.00	125.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.35	0.00	0.00	117.76	5,880,000.00 *gal*
Application event totals		2.35	0.00	0.00	117.76	

Field 1 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.47	0.00	0.00	122.09	6,270,000.00 *gal*
IW 5	Ground water	2.28	0.00	0.00	114.16	5,700,000.00 *gal*
Application event totals		4.75	0.00	0.00	236.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	1.51	0.00	0.00	74.54	3,828,000.00 *gal*
Application event totals		1.51	0.00	0.00	74.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	1.39	0.00	0.00	69.70	3,480,000.00 *gal*
Application event totals		1.39	0.00	0.00	69.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.10	12.31	215.19	1,353.29	3,132,000.00 *gal*
Application event totals		121.10	12.31	215.19	1,353.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.47	0.00	0.00	122.09	6,270,000.00 *gal*
IW 5	Ground water	2.28	0.00	0.00	114.16	5,700,000.00 *gal*
Application event totals		4.75	0.00	0.00	236.25	

Field 1 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	1.48	0.00	0.00	73.25	3,762,000.00 *gal*
IW 5	Ground water	1.37	0.00	0.00	68.50	3,420,000.00 *gal*
Application event totals		2.85	0.00	0.00	141.75	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.01	12.10	211.48	1,329.96	3,078,000.00 *gal*
Application event totals		119.01	12.10	211.48	1,329.96	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.39	0.00	0.00	118.23	6,072,000.00 *gal*
Application event totals		2.39	0.00	0.00	118.23	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.21	0.00	0.00	110.55	5,520,000.00 *gal*
Application event totals		2.21	0.00	0.00	110.55	

09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.29	0.00	0.00	113.09	5,808,000.00 *gal*
IW 5	Ground water	2.11	0.00	0.00	105.75	5,280,000.00 *gal*
Application event totals		4.40	0.00	0.00	218.84	

Field 2 - 11/11/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Field 2 - 11/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.82	18.46	161.69	1,542.24	2,484,000.00 *gal*
Application event totals		123.82	18.46	161.69	1,542.24	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.97	20.87	182.78	1,743.40	2,808,000.00 *gal*
Application event totals		139.97	20.87	182.78	1,743.40	

Field 2 - 06/26/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.59	0.00	0.00	128.00	5,478,000.00 *gal*
IW 5	Ground water	2.39	0.00	0.00	119.69	4,980,000.00 *gal*
Application event totals		4.98	0.00	0.00	247.69	

06/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Field 2 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	1.56	0.00	0.00	77.11	3,300,000.00 *gal*
IW 5	Ground water	1.44	0.00	0.00	72.10	3,000,000.00 *gal*
Application event totals		3.00	0.00	0.00	149.21	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	125.28	12.74	222.61	1,399.96	2,700,000.00 *gal*
Application event totals		125.28	12.74	222.61	1,399.96	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.50	0.00	0.00	123.37	5,280,000.00 *gal*
IW 5	Ground water	2.30	0.00	0.00	115.36	4,800,000.00 *gal*
Application event totals		4.80	0.00	0.00	238.73	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	1.50	0.00	0.00	74.02	3,168,000.00 *gal*
IW 5	Ground water	1.38	0.00	0.00	69.22	2,880,000.00 *gal*
Application event totals		2.88	0.00	0.00	143.24	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.26	12.23	213.71	1,343.96	2,592,000.00 *gal*
Application event totals		120.26	12.23	213.71	1,343.96	

Field 2 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.50	0.00	0.00	123.37	5,280,000.00 *gal*
IW 5	Ground water	2.30	0.00	0.00	115.36	4,800,000.00 *gal*
Application event totals		4.80	0.00	0.00	238.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.31	0.00	0.00	114.12	4,884,000.00 *gal*
IW 5	Ground water	2.13	0.00	0.00	106.71	4,440,000.00 *gal*
Application event totals		4.44	0.00	0.00	220.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.22	0.00	0.00	109.49	4,686,000.00 *gal*
IW 5	Ground water	2.04	0.00	0.00	102.38	4,260,000.00 *gal*
Application event totals		4.26	0.00	0.00	211.88	

Field 3 - 11/11/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	126.90	18.92	165.70	1,580.55	1,188,000.00 *gal*
Application event totals		126.90	18.92	165.70	1,580.55	

Field 3 - 11/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.43	20.64	180.77	1,724.24	1,296,000.00 *gal*
Application event totals		138.43	20.64	180.77	1,724.24	

Field 3 - 06/28/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.40	0.00	0.00	120.51	2,340,000.00 *gal*
IW 6	Ground water	2.58	0.00	0.00	126.43	2,574,000.00 *gal*
Application event totals		4.98	0.00	0.00	246.94	

06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	1.48	0.00	0.00	74.16	1,440,000.00 *gal*
IW 6	Ground water	1.59	0.00	0.00	77.80	1,584,000.00 *gal*
Application event totals		3.07	0.00	0.00	151.96	

Field 3 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	128.85	13.10	228.97	1,439.96	1,296,000.00 *gal*
Application event totals		128.85	13.10	228.97	1,439.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.16	0.00	0.00	108.15	2,100,000.00 *gal*
IW 6	Ground water	2.31	0.00	0.00	113.46	2,310,000.00 *gal*
Application event totals		4.47	0.00	0.00	221.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	118.12	12.01	209.89	1,319.96	1,188,000.00 *gal*
IW 5	Ground water	1.36	0.00	0.00	67.98	1,320,000.00 *gal*
IW 6	Ground water	1.45	0.00	0.00	71.32	1,452,000.00 *gal*
Application event totals		120.93	12.01	209.89	1,459.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.16	0.00	0.00	108.15	2,100,000.00 *gal*
IW 6	Ground water	2.31	0.00	0.00	113.46	2,310,000.00 *gal*
Application event totals		4.47	0.00	0.00	221.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.10	0.00	0.00	105.06	2,040,000.00 *gal*
IW 6	Ground water	2.25	0.00	0.00	110.22	2,244,000.00 *gal*
Application event totals		4.34	0.00	0.00	215.28	

Field 3 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.10	0.00	0.00	105.06	2,040,000.00 *gal*
IW 6	Ground water	2.25	0.00	0.00	110.22	2,244,000.00 *gal*
Application event totals		4.34	0.00	0.00	215.28	

Field 4 - 11/12/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	191.65	117.97	440.08	9,781.04	770.00 *ton*
Application event totals		191.65	117.97	440.08	9,781.04	

Field 4 - 06/30/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.28	0.00	0.00	114.05	3,480,000.00 *gal*
IW 6	Ground water	2.44	0.00	0.00	119.65	3,828,000.00 *gal*
Application event totals		4.72	0.00	0.00	233.70	

Field 4 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	1.49	0.00	0.00	74.72	2,280,000.00 *gal*
IW 6	Ground water	1.60	0.00	0.00	78.39	2,508,000.00 *gal*
Application event totals		3.09	0.00	0.00	153.11	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.83	13.20	230.71	1,450.86	2,052,000.00 *gal*
Application event totals		129.83	13.20	230.71	1,450.86	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.20	0.00	0.00	110.12	3,360,000.00 *gal*
IW 6	Ground water	2.36	0.00	0.00	115.52	3,696,000.00 *gal*
Application event totals		4.55	0.00	0.00	225.64	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	1.41	0.00	0.00	70.79	2,160,000.00 *gal*
IW 6	Ground water	1.51	0.00	0.00	74.26	2,376,000.00 *gal*
Application event totals		2.93	0.00	0.00	145.05	

Field 4 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.00	12.51	218.56	1,374.50	1,944,000.00 *gal*
Application event totals		123.00	12.51	218.56	1,374.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.16	0.00	0.00	108.15	3,300,000.00 *gal*
IW 6	Ground water	2.31	0.00	0.00	113.46	3,630,000.00 *gal*
Application event totals		4.47	0.00	0.00	221.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.16	0.00	0.00	108.15	3,300,000.00 *gal*
IW 6	Ground water	2.31	0.00	0.00	113.46	3,630,000.00 *gal*
Application event totals		4.47	0.00	0.00	221.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.12	0.00	0.00	106.18	3,240,000.00 *gal*
IW 6	Ground water	2.27	0.00	0.00	111.40	3,564,000.00 *gal*
Application event totals		4.39	0.00	0.00	217.58	

Field 5 - 11/12/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 5 - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	201.88	30.10	263.62	2,514.52	1,890,000.00 *gal*
Application event totals		201.88	30.10	263.62	2,514.52	

Field 5 - 06/28/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.34	0.00	0.00	117.42	2,280,000.00 *gal*
IW 7	Ground water	2.13	0.00	0.00	111.22	2,508,000.00 *gal*
Application event totals		4.47	0.00	0.00	228.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.28	0.00	0.00	114.33	2,220,000.00 *gal*
IW 7	Ground water	2.07	0.00	0.00	108.30	2,442,000.00 *gal*
Application event totals		4.35	0.00	0.00	222.63	

Field 5 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	1.42	0.00	0.00	71.07	1,380,000.00 *gal*
IW 7	Ground water	1.29	0.00	0.00	67.32	1,518,000.00 *gal*
Application event totals		2.71	0.00	0.00	138.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.49	12.56	219.43	1,379.96	1,242,000.00 *gal*
Application event totals		123.49	12.56	219.43	1,379.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.22	0.00	0.00	111.24	2,160,000.00 *gal*
IW 7	Ground water	2.02	0.00	0.00	105.37	2,376,000.00 *gal*
Application event totals		4.24	0.00	0.00	216.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.49	12.56	219.43	1,379.96	1,242,000.00 *gal*
IW 5	Ground water	1.42	0.00	0.00	71.07	1,380,000.00 *gal*
IW 7	Ground water	1.29	0.00	0.00	67.32	1,518,000.00 *gal*
Application event totals		126.19	12.56	219.43	1,518.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.16	0.00	0.00	108.15	2,100,000.00 *gal*
IW 7	Ground water	1.96	0.00	0.00	102.44	2,310,000.00 *gal*
Application event totals		4.12	0.00	0.00	210.59	

Field 5 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.03	0.00	0.00	101.97	1,980,000.00 *gal*
IW 7	Ground water	1.85	0.00	0.00	96.59	2,178,000.00 *gal*
Application event totals		3.88	0.00	0.00	198.56	

Field 6 - 11/13/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	189.88	116.88	436.02	9,690.90	860.00 *ton*
Application event totals		189.88	116.88	436.02	9,690.90	

Field 6 - 06/16/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	260.70	166.79	445.71	6,657.66	1,100.00 *ton*
Application event totals		260.70	166.79	445.71	6,657.66	

Field 6 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.33	0.00	0.00	116.87	4,020,000.00 *gal*
IW 7	Ground water	2.12	0.00	0.00	110.70	4,422,000.00 *gal*
Application event totals		4.45	0.00	0.00	227.58	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.26	0.00	0.00	113.38	3,900,000.00 *gal*
IW 7	Ground water	2.06	0.00	0.00	107.40	4,290,000.00 *gal*
Application event totals		4.32	0.00	0.00	220.78	

07/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.23	0.00	0.00	111.64	3,840,000.00 *gal*
IW 7	Ground water	2.02	0.00	0.00	105.75	4,224,000.00 *gal*
Application event totals		4.25	0.00	0.00	217.39	

Field 6 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.23	0.00	0.00	111.64	3,840,000.00 *gal*
IW 7	Ground water	2.02	0.00	0.00	105.75	4,224,000.00 *gal*
Application event totals		4.25	0.00	0.00	217.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.16	0.00	0.00	108.15	3,720,000.00 *gal*
IW 7	Ground water	1.96	0.00	0.00	102.44	4,092,000.00 *gal*
Application event totals		4.12	0.00	0.00	210.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.09	0.00	0.00	104.66	3,600,000.00 *gal*
IW 7	Ground water	1.90	0.00	0.00	99.14	3,960,000.00 *gal*
Application event totals		3.99	0.00	0.00	203.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	2.02	0.00	0.00	101.17	3,480,000.00 *gal*
IW 7	Ground water	1.83	0.00	0.00	95.83	3,828,000.00 *gal*
Application event totals		3.85	0.00	0.00	197.01	

Field 7 - 11/13/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 7 - 11/13/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	189.26	28.22	247.14	2,357.36	1,620,000.00 *gal*
Application event totals		189.26	28.22	247.14	2,357.36	

Field 7 - 06/26/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	266.33	170.39	455.34	6,801.41	580.00 *ton*
Application event totals		266.33	170.39	455.34	6,801.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.49	0.00	0.00	122.89	2,244,000.00 *gal*
IW 7	Ground water	2.08	0.00	0.00	108.85	2,244,000.00 *gal*
Application event totals		4.57	0.00	0.00	231.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Field 7 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.34	0.00	0.00	115.66	2,112,000.00 *gal*
IW 7	Ground water	1.96	0.00	0.00	102.44	2,112,000.00 *gal*
Application event totals		4.30	0.00	0.00	218.10	

07/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.27	0.00	0.00	112.05	2,046,000.00 *gal*
IW 7	Ground water	1.90	0.00	0.00	99.24	2,046,000.00 *gal*
Application event totals		4.17	0.00	0.00	211.29	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.34	0.00	0.00	115.66	2,112,000.00 *gal*
IW 7	Ground water	1.96	0.00	0.00	102.44	2,112,000.00 *gal*
Application event totals		4.30	0.00	0.00	218.10	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.27	0.00	0.00	112.05	2,046,000.00 *gal*
IW 7	Ground water	1.90	0.00	0.00	99.24	2,046,000.00 *gal*
Application event totals		4.17	0.00	0.00	211.29	

Field 7 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.19	0.00	0.00	108.43	1,980,000.00 *gal*
IW 7	Ground water	1.84	0.00	0.00	96.04	1,980,000.00 *gal*
Application event totals		4.03	0.00	0.00	204.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	2.12	0.00	0.00	104.82	1,914,000.00 *gal*
IW 7	Ground water	1.78	0.00	0.00	92.84	1,914,000.00 *gal*
Application event totals		3.90	0.00	0.00	197.66	

Field 8 - 11/13/2022: Oats, silage-soft dough

Field name: Field 8

Crop: Oats, silage-soft dough Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	187.59	115.47	430.76	9,574.04	370.00 *ton*
Application event totals		187.59	115.47	430.76	9,574.04	

Field 8 - 06/28/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 8 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	261.23	167.13	446.61	6,671.11	480.00 *ton*
Application event totals		261.23	167.13	446.61	6,671.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	4.37	0.00	0.00	228.17	3,969,000.00 *gal*
Application event totals		4.37	0.00	0.00	228.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.85	0.00	0.00	201.09	3,498,000.00 *gal*
Application event totals		3.85	0.00	0.00	201.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.92	0.00	0.00	204.89	3,564,000.00 *gal*
Application event totals		3.92	0.00	0.00	204.89	

Field 8 - 06/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.78	0.00	0.00	197.30	3,432,000.00 *gal*
Application event totals		3.78	0.00	0.00	197.30	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.78	0.00	0.00	197.30	3,432,000.00 *gal*
Application event totals		3.78	0.00	0.00	197.30	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.70	0.00	0.00	193.50	3,366,000.00 *gal*
Application event totals		3.70	0.00	0.00	193.50	

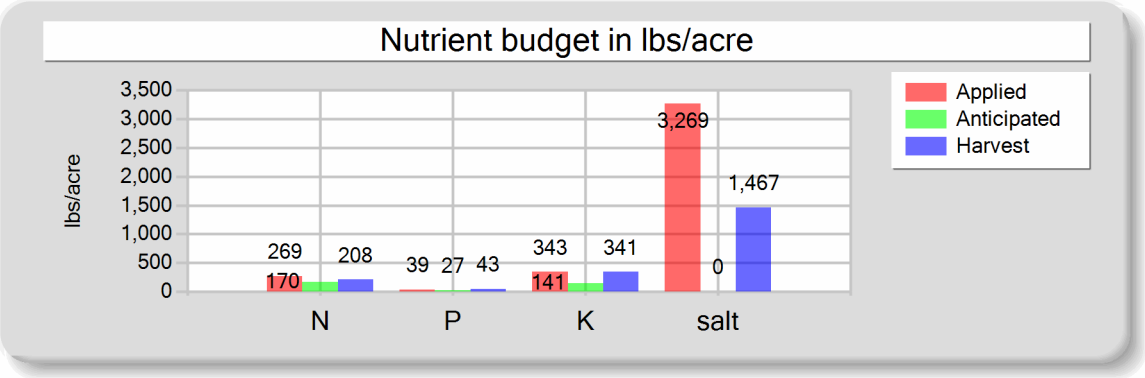
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.70	0.00	0.00	193.50	3,366,000.00 *gal*
Application event totals		3.70	0.00	0.00	193.50	

B. NUTRIENT BUDGET

Field 1 - 11/10/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.45	39.13	342.70	3,268.87
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.45	39.13	342.70	3,268.87
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	207.95	43.30	341.12	1,467.31
Nutrient balance	61.49	-4.16	1.58	1,801.56
Applied to removed ratio	1.30	0.90	1.00	2.23

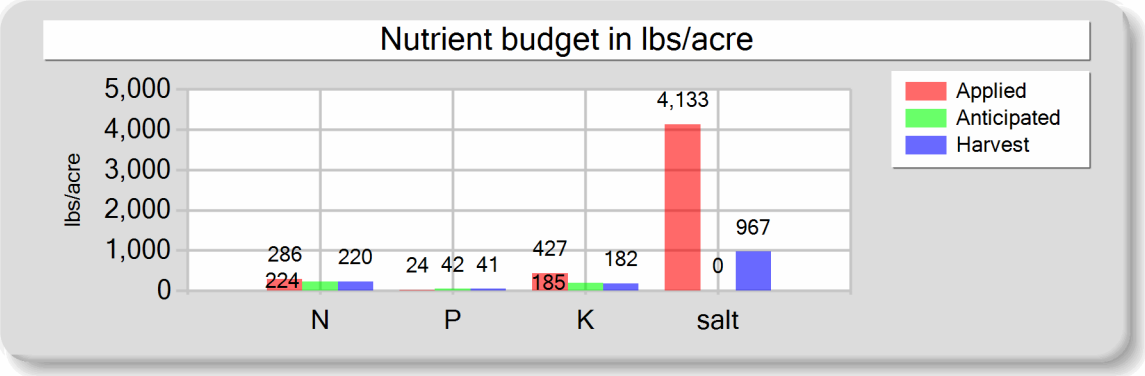
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
6,318,000.00	*gallons*
232.67	*acre-inches*
2.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/22/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.11	24.41	426.67	2,683.25
Fresh water	29.14	0.00	0.00	1,449.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.25	24.41	426.67	4,133.06
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	220.00	40.70	182.05	966.94
Nutrient balance	66.25	-16.29	244.62	3,166.12
Applied to removed ratio	1.30	0.60	2.34	4.27

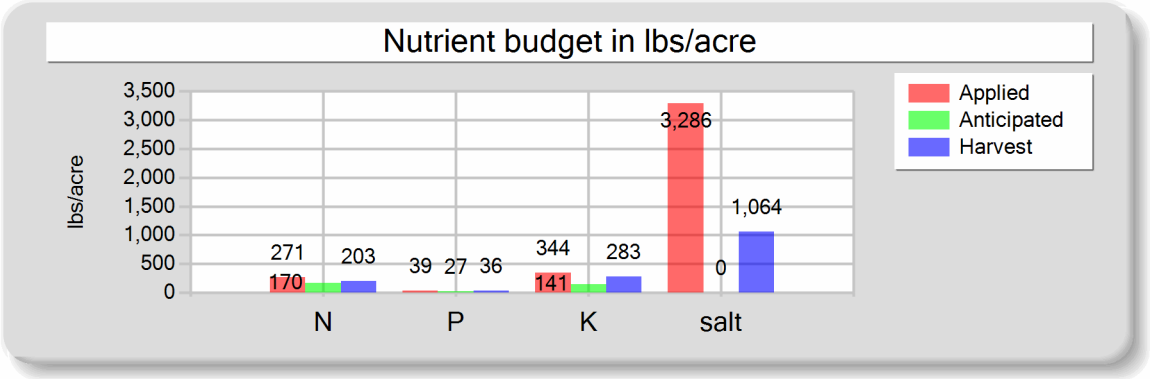
Fresh water applied	
73,458,000.00	*gallons*
2,705.21	*acre-inches*
30.06	*inches/acre*

Process wastewater applied	
6,210,000.00	*gallons*
228.69	*acre-inches*
2.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/11/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.79	39.33	344.46	3,285.63
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.79	39.33	344.46	3,285.63
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	203.44	35.67	282.90	1,063.90
Nutrient balance	67.35	3.66	61.56	2,221.73
Applied to removed ratio	1.33	1.10	1.22	3.09

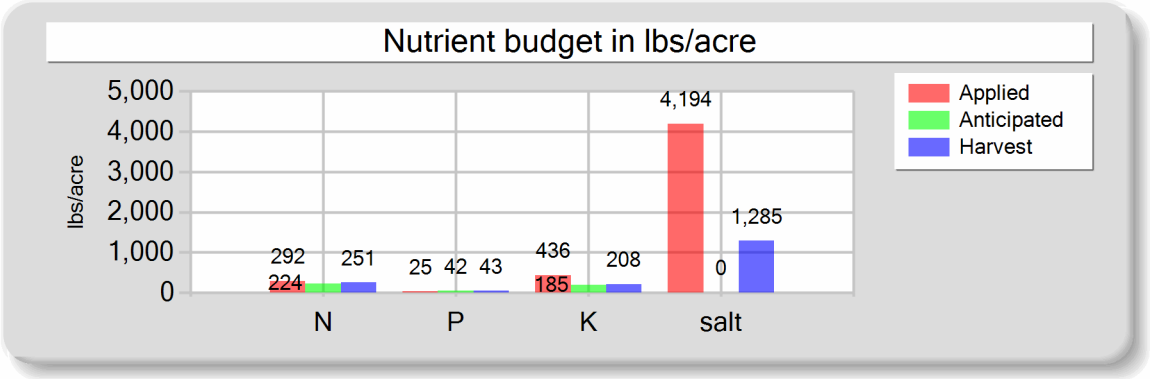
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
5,292,000.00	*gallons*
194.89	*acre-inches*
2.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/26/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	245.54	24.97	436.32	2,743.92
Fresh water	29.15	0.00	0.00	1,450.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.69	24.97	436.32	4,194.22
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	251.16	43.16	208.00	1,284.50
Nutrient balance	40.53	-18.19	228.32	2,909.72
Applied to removed ratio	1.16	0.58	2.10	3.27

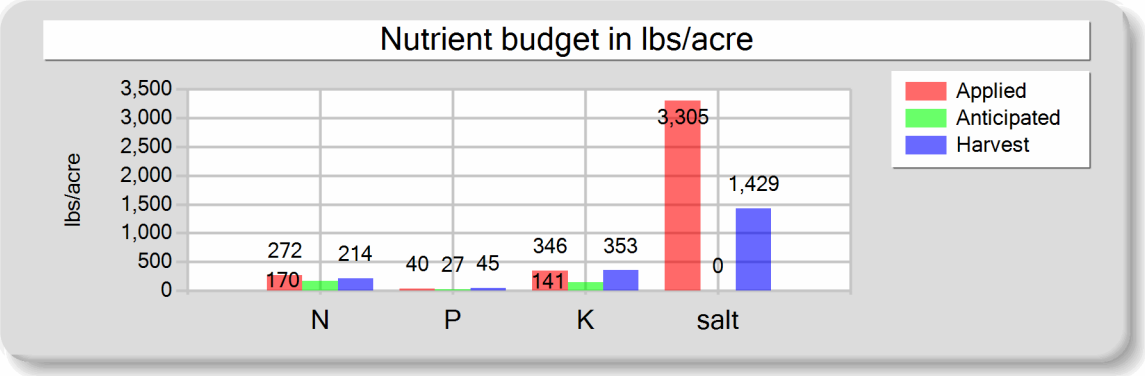
Fresh water applied	
61,236,000.00	*gallons*
2,255.11	*acre-inches*
30.07	*inches/acre*

Process wastewater applied	
5,292,000.00	*gallons*
194.89	*acre-inches*
2.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/11/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.33	39.56	346.47	3,304.79
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.33	39.56	346.47	3,304.79
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	214.34	44.62	352.56	1,428.90
Nutrient balance	57.99	-5.05	-6.09	1,875.89
Applied to removed ratio	1.27	0.89	0.98	2.31

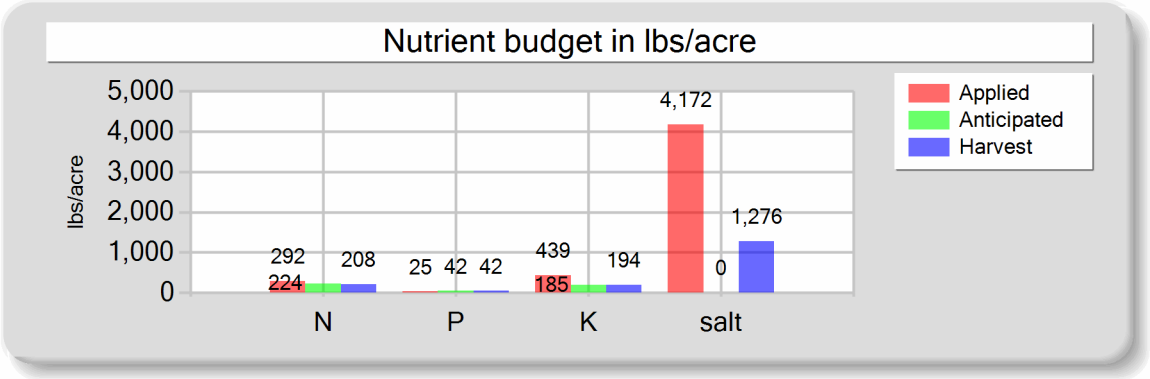
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,484,000.00	*gallons*
91.48	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/28/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.97	25.11	438.86	2,759.92
Fresh water	28.49	0.00	0.00	1,411.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.46	25.11	438.86	4,171.89
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	207.58	41.85	193.63	1,275.99
Nutrient balance	84.88	-16.74	245.24	2,895.90
Applied to removed ratio	1.41	0.60	2.27	3.27

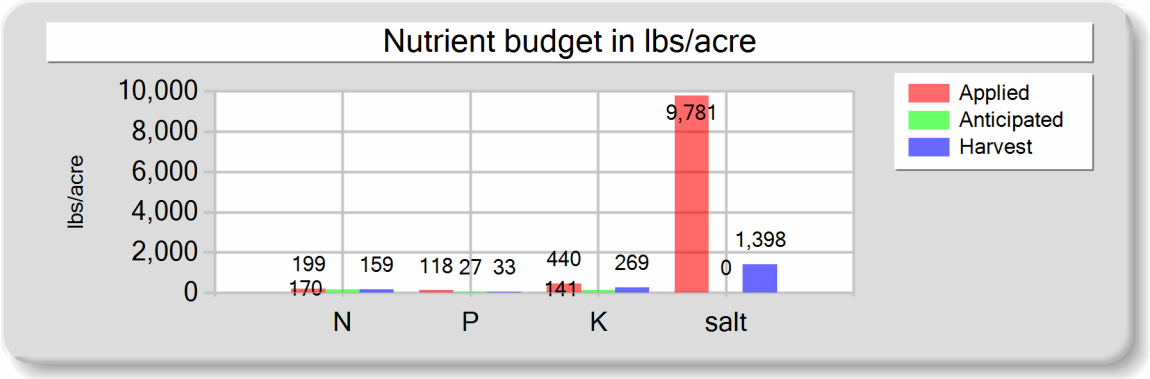
Fresh water applied	
28,098,000.00	*gallons*
1,034.75	*acre-inches*
29.56	*inches/acre*

Process wastewater applied	
2,484,000.00	*gallons*
91.48	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/12/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	191.65	117.97	440.08	9,781.04
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.65	117.97	440.08	9,781.04
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	158.65	32.77	268.94	1,398.49
Nutrient balance	39.99	85.20	171.14	8,382.56
Applied to removed ratio	1.25	3.60	1.64	6.99

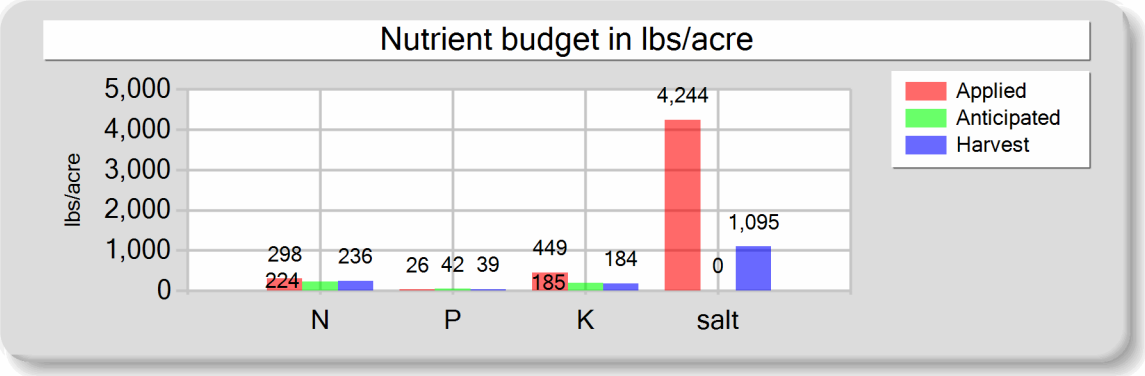
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 4 - 06/30/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.83	25.71	449.27	2,825.37
Fresh water	28.62	0.00	0.00	1,418.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.44	25.71	449.27	4,243.67
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	235.93	39.06	184.01	1,095.19
Nutrient balance	62.52	-13.36	265.26	3,148.48
Applied to removed ratio	1.26	0.66	2.44	3.87

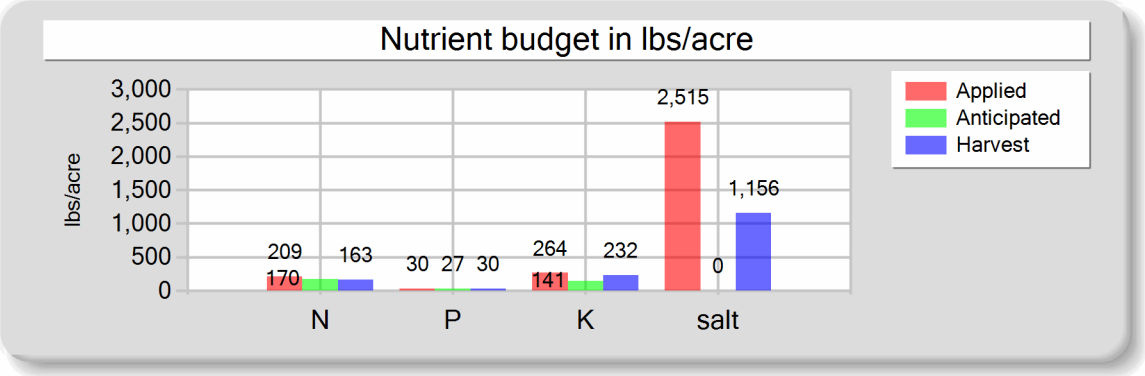
Fresh water applied	
44,352,000.00	*gallons*
1,633.33	*acre-inches*
29.70	*inches/acre*

Process wastewater applied	
3,996,000.00	*gallons*
147.16	*acre-inches*
2.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/12/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.88	30.10	263.62	2,514.52
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.88	30.10	263.62	2,514.52
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	163.33	29.92	232.32	1,156.28
Nutrient balance	45.55	0.18	31.30	1,358.23
Applied to removed ratio	1.28	1.01	1.13	2.17

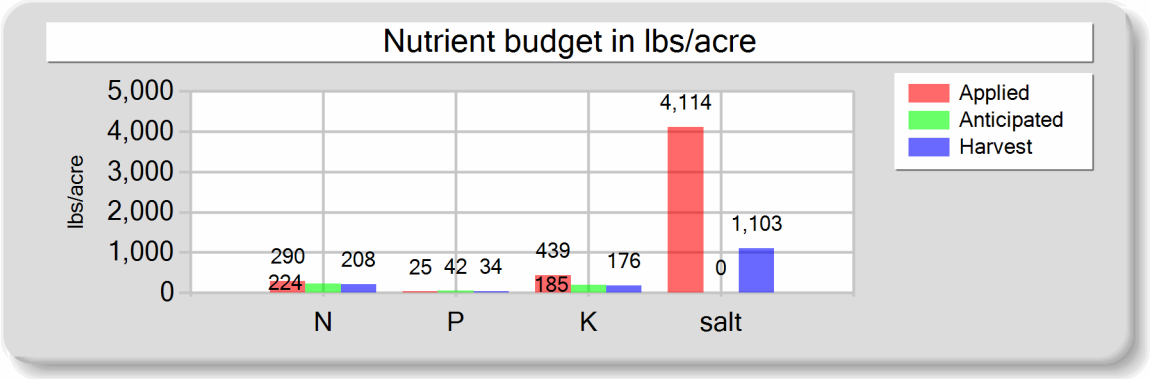
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,890,000.00	*gallons*
69.60	*acre-inches*
1.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 06/28/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.97	25.11	438.86	2,759.92
Fresh water	26.48	0.00	0.00	1,353.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.45	25.11	438.86	4,113.74
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	207.64	33.75	176.18	1,102.61
Nutrient balance	82.81	-8.64	262.69	3,011.13
Applied to removed ratio	1.40	0.74	2.49	3.73

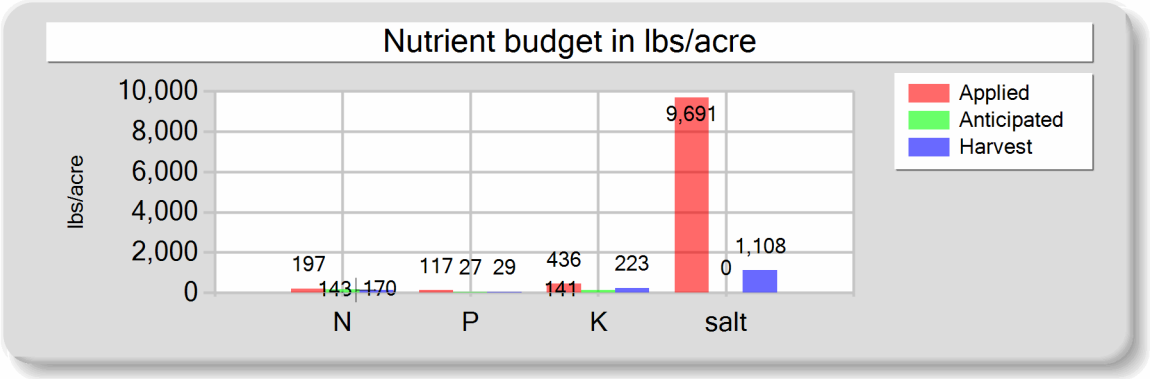
Fresh water applied	
28,350,000.00	*gallons*
1,044.03	*acre-inches*
29.83	*inches/acre*

Process wastewater applied	
2,484,000.00	*gallons*
91.48	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/13/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	189.88	116.88	436.02	9,690.90
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.88	116.88	436.02	9,690.90
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	142.76	28.84	222.86	1,107.57
Nutrient balance	54.12	88.05	213.16	8,583.32
Applied to removed ratio	1.38	4.05	1.96	8.75

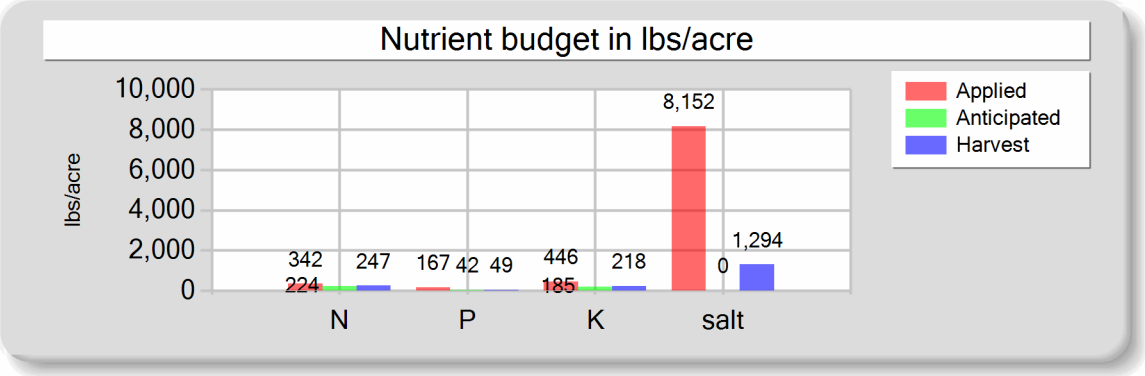
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 6 - 06/16/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	260.70	166.79	445.71	6,657.66
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	29.23	0.00	0.00	1,494.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	342.13	166.79	445.71	8,152.20
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	247.10	49.31	217.62	1,293.71
Nutrient balance	95.03	117.48	228.10	6,858.49
Applied to removed ratio	1.38	3.38	2.05	6.30

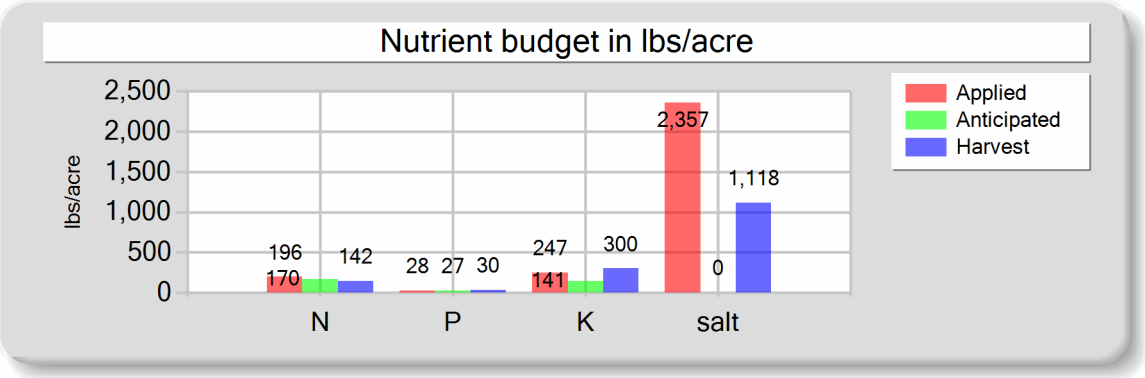
Fresh water applied	
55,440,000.00	*gallons*
2,041.67	*acre-inches*
32.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 11/13/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.26	28.22	247.14	2,357.36
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.26	28.22	247.14	2,357.36
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	141.77	30.29	300.50	1,117.74
Nutrient balance	54.50	-2.07	-53.35	1,239.62
Applied to removed ratio	1.38	0.93	0.82	2.11

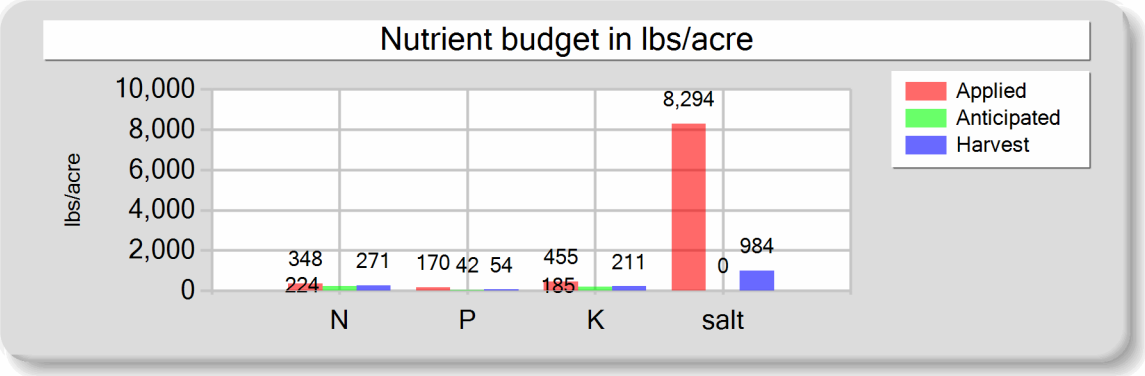
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,620,000.00	*gallons*
59.66	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 06/26/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	266.33	170.39	455.34	6,801.41
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	29.44	0.00	0.00	1,492.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	347.97	170.39	455.34	8,294.06
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	271.08	54.22	210.84	984.48
Nutrient balance	76.89	116.18	244.50	7,309.58
Applied to removed ratio	1.28	3.14	2.16	8.42

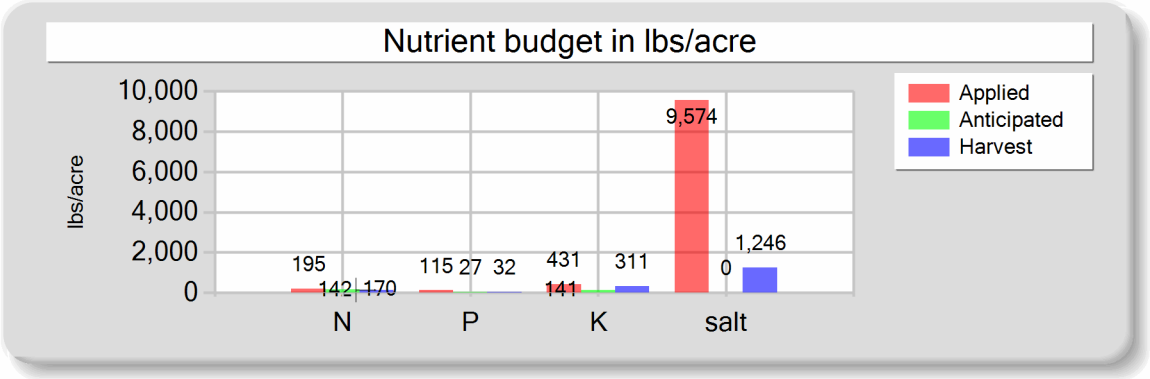
Fresh water applied	
28,908,000.00	*gallons*
1,064.58	*acre-inches*
33.27	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 11/13/2022: Oats, silage-soft dough

Field name: Field 8 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	187.59	115.47	430.76	9,574.04
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.59	115.47	430.76	9,574.04
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	142.13	31.75	311.47	1,246.15
Nutrient balance	52.46	83.72	119.29	8,327.89
Applied to removed ratio	1.37	3.64	1.38	7.68

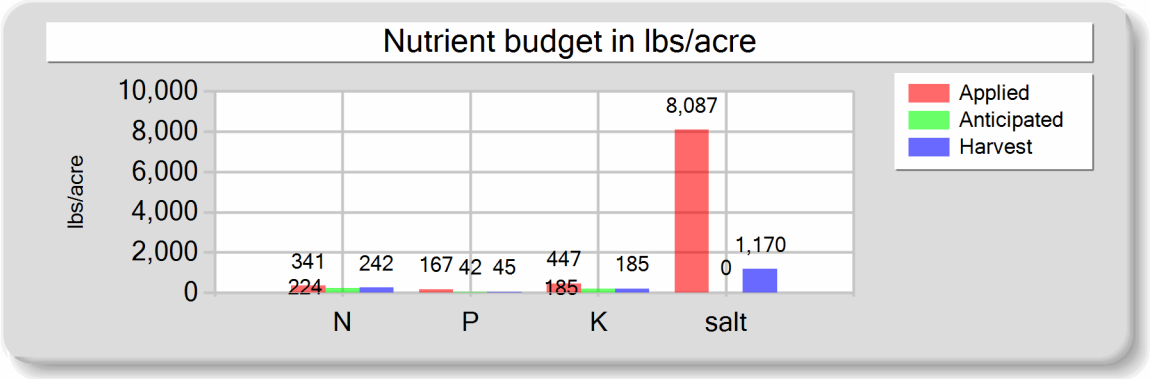
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 8 - 06/28/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	45.20	0.00	0.00	0.00
Dry manure	261.23	167.13	446.61	6,671.11
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	27.10	0.00	0.00	1,415.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	340.52	167.13	446.61	8,086.86
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	242.35	45.41	185.33	1,169.78
Nutrient balance	98.17	121.72	261.29	6,917.08
Applied to removed ratio	1.41	3.68	2.41	6.91

Fresh water applied	
24,627,000.00	*gallons*
906.93	*acre-inches*
33.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	8,310.00	31,000.00							68.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	11,900.00	31,800.00							47.50
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,300.00	6,420.00	44,600.00							41.60
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/24/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	483.00	378.00	0.00	0.00	58.50	929.00								7,440.00	6,340
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	448.00	291.00	0.00	0.00	66.80	585.00	149.00	108.00	163.00	3,140.00	0.00	253.00	342.00	6,770.00	5,580
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	50.00	1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	437.00	274.00	0.00	0.00	88.20	932.00								6,450.00	7,920
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	417.00	336.00	0.00	0.00	42.40	741.00								6,580.00	4,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	454.00	316.00	0.00	0.00	44.40	798.00								6,490.00	7,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW 4

Fresh Water

Sample description: Fresh Water

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.25	0.00	4.25								328.00	210
DL	0.50	0.50	0.50								1.00	10

IW 5

Fresh Water

Sample description: Fresh Water

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.31	0.00	4.31								322.00	216
DL	0.50	0.50	0.50								1.00	10

IW 6

IW 6

Fresh Water

Sample description: Fresh Water

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.20	0.00	4.20								329.00	206
DL	0.50	0.50	0.50								1.00	10

IW 7

Fresh Water

Sample description: Fresh Water

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.56	0.00	3.50								298.00	186
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

Field 1

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.00				
DL			4.00				

Field 2

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			100.00				
DL			4.00				

Field 3

Fresh Water

Sample and source description: Fresh Water

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			100.00				
DL			4.00				

Field 4

Fresh Water

Sample and source description: Fresh Water

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			117.00				
DL			4.00				

Field 5

Field 5

Soils

Sample and source description: Soils

Sample date: 01/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			117.00				
DL			4.00				

Field 6

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			4.00				

Field 7

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			75.00				
DL			4.00				

Field 8

Field 8

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			68.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 11/10/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,340.00	1,320.00	10,400.00		11.50
DL	100.00	100.00	100.00		0.01

Field 1 - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	740.00	3,310.00		5.36
DL	100.00	100.00	100.00		0.01

Field 2 - 11/11/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,270.00	1,450.00	11,500.00		10.20
DL	100.00	100.00	100.00		0.01

Field 2 - 06/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,830.00	830.00	4,000.00		6.90
DL	100.00	100.00	100.00		0.01

Field 3 - 11/11/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,870.00	1,430.00	11,300.00		10.20
DL	100.00	100.00	100.00		0.01

Field 3 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,720.00	750.00	3,470.00		7.94
DL	100.00	100.00	100.00		0.01

Field 4 - 11/12/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 48.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,020.00	1,450.00	11,900.00		11.90
DL	100.00	100.00	100.00		0.01

Field 4 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,590.00	760.00	3,580.00		5.52
DL	100.00	100.00	100.00		0.01

Field 5 - 11/12/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,640.00	850.00	6,600.00		10.70
DL	100.00	100.00	100.00		0.01

Field 5 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	590.00	3,080.00		6.74
DL	100.00	100.00	100.00		0.01

Field 6 - 11/13/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,010.00	810.00	6,260.00		8.06
DL	100.00	100.00	100.00		0.01

Field 6 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,610.00	920.00	4,060.00		7.27
DL	100.00	100.00	100.00		0.01

Field 7 - 11/13/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	750.00	7,440.00		10.10
DL	100.00	100.00	100.00		0.01

Field 7 - 06/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,400.00	1,080.00	4,200.00		4.76
DL	100.00	100.00	100.00		0.01

Field 8 - 11/13/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,760.00	840.00	8,240.00		9.90
DL	100.00	100.00	100.00		0.01

Field 8 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,590.00	860.00	3,510.00		6.33
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

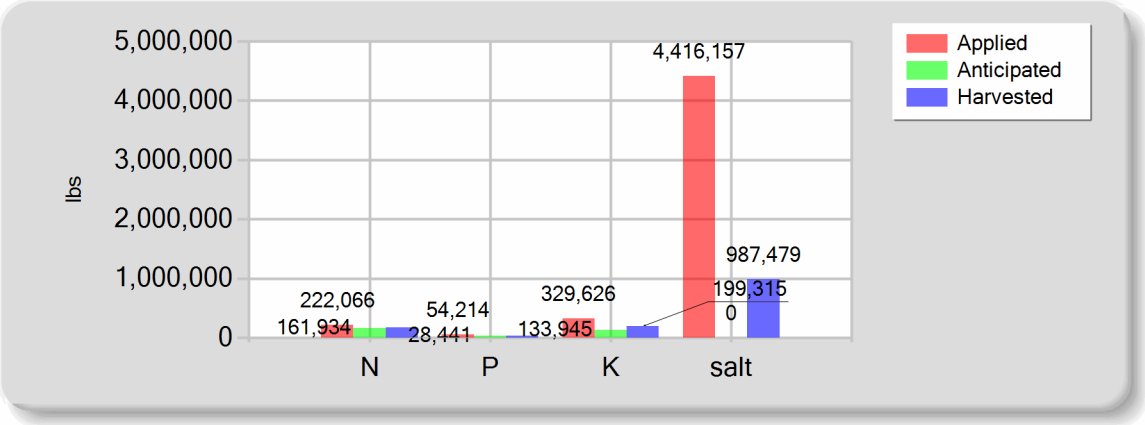
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

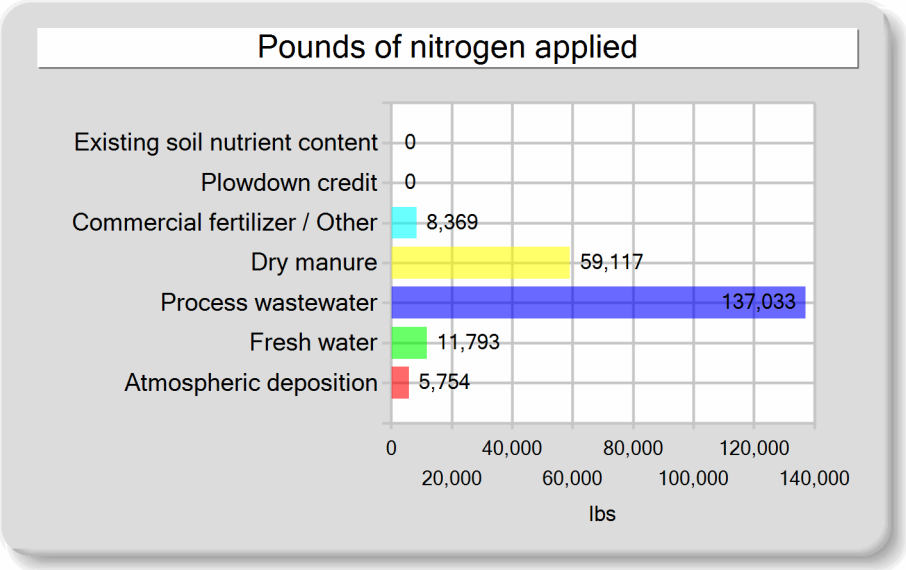
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

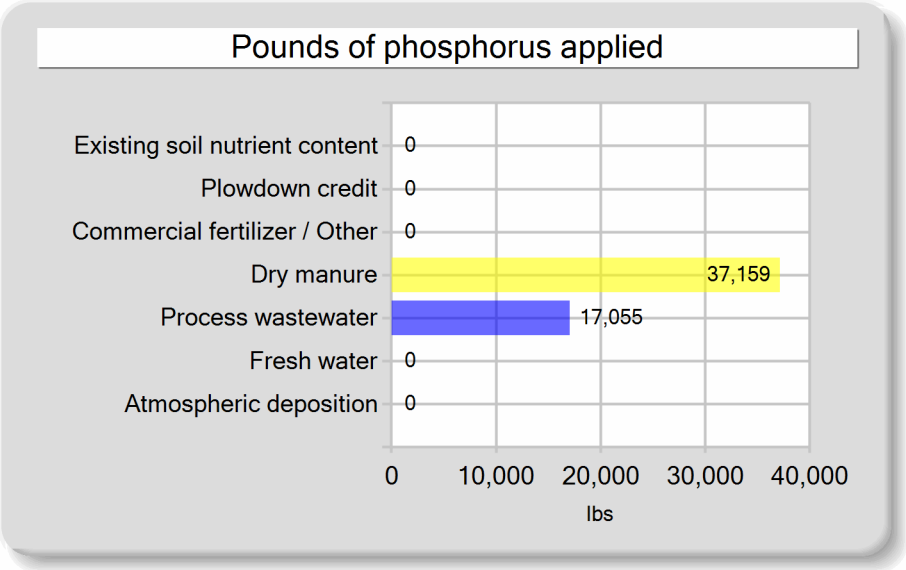
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8,369.20	0.00	0.00	0.00
Dry manure	59,117.04	37,158.84	117,131.52	2,207,832.00
Process wastewater	137,032.53	17,054.72	212,494.13	1,615,607.69
Fresh water	11,792.80	0.00	0.00	592,717.19
Atmospheric deposition	5,754.00	0.00	0.00	0.00
Total nutrients applied	222,065.57	54,213.56	329,625.65	4,416,156.88
Anticipated crop nutrient removal	161,934.00	28,441.20	133,944.90	0.00
Actual crop nutrient removal	169,826.44	32,310.20	199,314.68	987,478.58
Nutrient balance	52,239.13	21,903.36	130,310.97	3,428,678.30
Applied to removed ratio	1.31	1.68	1.65	4.47

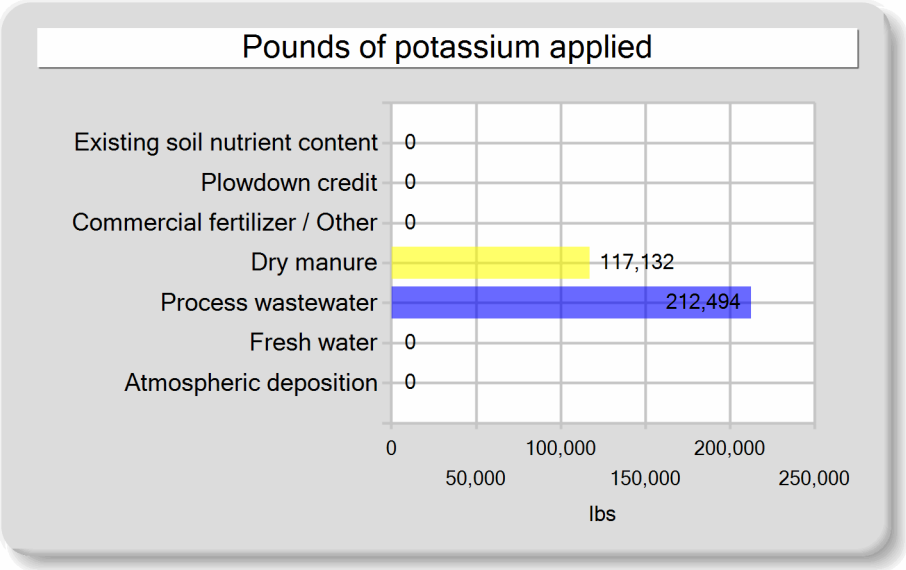
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

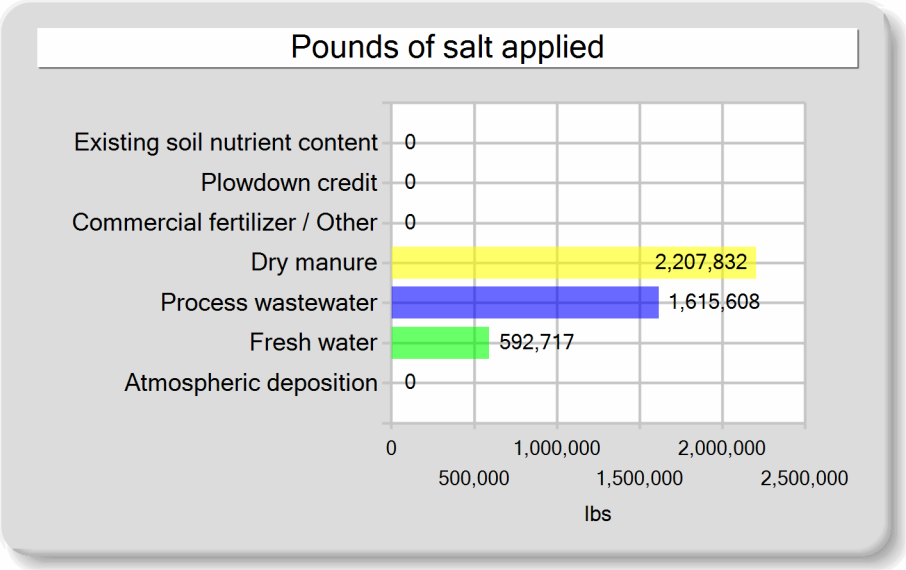


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David Koolhaas (May 22, 2024 16:51 PDT)	David Koolhaas (May 22, 2024 16:51 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Dodds Road LP Vanderschaaf	David Koolhaas
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 22, 2024	May 22, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 13:09

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0938-01	IW 4	Canal Water			11/15/2023 8:00
23K0938-02	IW 5	Canal Water			11/15/2023 8:00
23K0938-03	IW 6	Canal Water			11/15/2023 8:00
23K0938-04	IW 7	Canal Water			11/15/2023 8:00

Default Cooler Temperature on Receipt °C: 0.6

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 13:09

Sample Results

Sample: IW 4
23K0938-01 (Water)

Sampled: 11/15/2023 8:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.33**	mmhos/cm	0.01	1		11/27/23 10:04	SM 2510 B		BEK0619
Electrical Conductivity umhos	**328**	umhos/cm	10.0	1		11/27/23 10:04	SM 2510 B		BEK0619
Nitrate Nitrogen as NO3N	**4.2**	mg/L	0.1	1	10	11/17/23 01:01	EPA 300.0		BEK0572
Temperature	**25.0**	units	0.0	1		11/27/23 10:04	SM 4500-H+	H	BEK0619
pH	**8.1**	units	1.0	1		11/27/23 10:04	SM 4500-H+	H	BEK0619
Total Filterable Solids (TDS)	**210**	mg/L	10.0	1		11/20/23 14:52	SM 2540 C		BEK0609
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/21/23 09:29	SM 4500-NH3 C		BEK0631
Total Nitrogen	**4.25**	mg/L	1.00	1		11/21/23 09:29	SM 4500-NH3 C		BEK0631

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DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 13:09

Sample Results
(Continued)

Sample: IW 5
23K0938-02 (Water)

Sampled: 11/15/2023 8:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.32**	mmhos/cm	0.01	1		11/27/23 10:06	SM 2510 B		BEK0619
Electrical Conductivity umhos	**322**	umhos/cm	10.0	1		11/27/23 10:06	SM 2510 B		BEK0619
Nitrate Nitrogen as NO3N	**4.0**	mg/L	0.1	1	10	11/17/23 01:22	EPA 300.0		BEK0572
Temperature	**25.0**	units	0.0	1		11/27/23 10:06	SM 4500-H+	H	BEK0619
pH	**8.2**	units	1.0	1		11/27/23 10:06	SM 4500-H+	H	BEK0619
Total Filterable Solids (TDS)	**216**	mg/L	10.0	1		11/20/23 14:52	SM 2540 C		BEK0609
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/21/23 09:30	SM 4500-NH3 C		BEK0631
Total Nitrogen	**4.31**	mg/L	1.00	1		11/21/23 09:30	SM 4500-NH3 C		BEK0631

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DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 13:09

Sample Results
(Continued)

Sample: IW 6
23K0938-03 (Water)

Sampled: 11/15/2023 8:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.33**	mmhos/cm	0.01	1		11/27/23 10:07	SM 2510 B		BEK0619
Electrical Conductivity umhos	**329**	umhos/cm	10.0	1		11/27/23 10:07	SM 2510 B		BEK0619
Nitrate Nitrogen as NO3N	**4.2**	mg/L	0.1	1	10	11/17/23 01:43	EPA 300.0		BEK0572
Temperature	**25.0**	units	0.0	1		11/27/23 10:07	SM 4500-H+	H	BEK0619
pH	**8.2**	units	1.0	1		11/27/23 10:07	SM 4500-H+	H	BEK0619
Total Filterable Solids (TDS)	**206**	mg/L	10.0	1		11/20/23 14:52	SM 2540 C		BEK0609
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/21/23 09:32	SM 4500-NH3 C		BEK0631
Total Nitrogen	**4.20**	mg/L	1.00	1		11/21/23 09:32	SM 4500-NH3 C		BEK0631

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DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 13:09

Sample Results
(Continued)

Sample: IW 7
23K0938-04 (Water)

Sampled: 11/15/2023 8:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.30**	mmhos/cm	0.01	1		11/27/23 10:08	SM 2510 B		BEK0619
Electrical Conductivity umhos	**298**	umhos/cm	10.0	1		11/27/23 10:08	SM 2510 B		BEK0619
Nitrate Nitrogen as NO3N	**3.5**	mg/L	0.1	1	10	11/17/23 02:04	EPA 300.0		BEK0572
Temperature	**25.0**	units	0.0	1		11/27/23 10:08	SM 4500-H+	H	BEK0619
pH	**8.1**	units	1.0	1		11/27/23 10:08	SM 4500-H+	H	BEK0619
Total Filterable Solids (TDS)	**186**	mg/L	10.0	1		11/20/23 14:52	SM 2540 C		BEK0609
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/21/23 09:33	SM 4500-NH3 C		BEK0631
Total Nitrogen	**3.56**	mg/L	1.00	1		11/21/23 09:33	SM 4500-NH3 C		BEK0631

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WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

26311

Purchase Order No

Bill To: 0 Acct # 13 Cons #

No. Samples: 4 No of Bottles: 8

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[✓] Ag Water [ ] Groundwater [ ] Monitoring Well
AWS 11/16/23
Other:

Name: Koolhaas Dairy
Address: 26805 Dobbs Rd
City: Escalon State: CA Zip: 95320
Telephone: 209 606-3424 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCW1 +TN
AWS 11/16/23

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW 4	11/15/23	Started 8 AM	0.6	
2	5			6.8	
3	6			2.8	
4	7			3.4	
5					
6					
7					
8					
9			JG		
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

11/16/23 15:40 23K0938

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	11-16-23 11:00	11-16-23 3:25
Third	JMP	DLV	11/16/23 15:40	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 12:22

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0936-01	DW Dairy Left	Well Water			11/15/2023 7:30
23K0936-02	DW Dairy Right	Well Water			11/15/2023 7:30
23K0936-03	DW House	Well Water			11/15/2023 7:30

Default Cooler Temperature on Receipt °C: 6.6

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 12:22

Sample Results

Sample: DW Dairy Left
23K0936-01 (Water)

Sampled: 11/15/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.36**	mmhos/cm	0.01	1		11/27/23 09:52	SM 2510 B		BEK0619
Electrical Conductivity umhos	**361**	umhos/cm	10.0	1		11/27/23 09:52	SM 2510 B		BEK0619
Ammonia (as N)	*	mg/L	0.00	1		11/17/23 11:24	Field		BEK0635
Nitrate Nitrogen as NO3N	**5.1**	mg/L	0.1	1	10	11/17/23 07:41	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/27/23 09:52	SM 4500-H+	H	BEK0619
pH	**8.1**	units	1.0	1		11/27/23 09:52	SM 4500-H+	H	BEK0619
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/21/23 09:25	SM 4500-NH3 C		BEK0631
Total Nitrogen	**5.22**	mg/L	1.00	1		11/21/23 09:25	SM 4500-NH3 C		BEK0631

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DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 12:22

Sample Results
(Continued)

Sample: DW Dairy Right
23K0936-02 (Water)

Sampled: 11/15/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.34**	mmhos/cm	0.01	1		11/27/23 09:59	SM 2510 B		BEK0619
Electrical Conductivity umhos	**344**	umhos/cm	10.0	1		11/27/23 09:59	SM 2510 B		BEK0619
Ammonia (as N)	*	mg/L	0.00	1		11/17/23 11:24	Field		BEK0635
Nitrate Nitrogen as NO3N	**4.8**	mg/L	0.1	1	10	11/17/23 08:02	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/27/23 09:59	SM 4500-H+	H	BEK0619
pH	**8.0**	units	1.0	1		11/27/23 09:59	SM 4500-H+	H	BEK0619
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/21/23 09:26	SM 4500-NH3 C		BEK0631
Total Nitrogen	**5.30**	mg/L	1.00	1		11/21/23 09:26	SM 4500-NH3 C		BEK0631

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DELLAVALLE™
LABORATORY INC

Koolhaas Dairy
26805 Dodds Rd
Escalon, CA 95320

Account# 00-0026311
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 11/16/2023 15:40
Reported: 11/29/2023 12:22

Sample Results
(Continued)

Sample: DW House
23K0936-03 (Water)

Sampled: 11/15/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.32**	mmhos/cm	0.01	1		11/27/23 10:00	SM 2510 B		BEK0619
Electrical Conductivity umhos	**323**	umhos/cm	10.0	1		11/27/23 10:00	SM 2510 B		BEK0619
Ammonia (as N)	*	mg/L	0.00	1		11/17/23 11:24	Field		BEK0635
Nitrate Nitrogen as NO3N	**4.2**	mg/L	0.1	1	10	11/17/23 08:22	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/27/23 10:00	SM 4500-H+	H	BEK0619
pH	**8.1**	units	1.0	1		11/27/23 10:00	SM 4500-H+	H	BEK0619
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		11/21/23 09:28	SM 4500-NH3 C		BEK0631
Total Nitrogen	**4.65**	mg/L	1.00	1		11/21/23 09:28	SM 4500-NH3 C		BEK0631

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WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

26311

Purchase Order No | Bill To: Acct # 0 | Cons # 13

No. Samples: 3 | No of Bottles: 6

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Koolhaas Dairy
Address: 26805 Dobbs Rd
City: Escalon State: CA Zip: 95320
Telephone: 209 606-3424 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (Please indicate Analysis)

DWW1 + TN

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	DW Dairy Left	11/15/23	Started 7:30 AM	6.6	
2	↓ Dairy Right	↓		8.3	
3	↓ House			0.9	
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

11/16/23 15:40 23K0936

JG

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	11-16-23 11:00	11-16-23 3:25
Third	JMD	DLI	11/16/23 15:40	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

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Invoicing Information:

Sampling hrs
Miles
Consulting

Shipping

$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

2023 Annual Report Koolhaas

Final Audit Report 2024-05-22

Created:	2024-05-22
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAACnG7aZ1eNWNj-Qff_7UoMhvStvAjW_co

"2023 Annual Report Koolhaas" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-05-22 - 8:33:19 PM GMT

Document emailed to koolhaasdairy@gmail.com for signature
2024-05-22 - 8:34:40 PM GMT

Email viewed by koolhaasdairy@gmail.com
2024-05-22 - 11:50:23 PM GMT

Signer koolhaasdairy@gmail.com entered name at signing as David Koolhaas
2024-05-22 - 11:51:41 PM GMT

Document e-signed by David Koolhaas (koolhaasdairy@gmail.com)
Signature Date: 2024-05-22 - 11:51:43 PM GMT - Time Source: server

Agreement completed.
2024-05-22 - 11:51:43 PM GMT